

MEVIN JOSE

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Summary

AI/ML undergraduate specializing in reliable LLM systems, evaluation-driven ML, and distributed AI infrastructure. Authored a Text-to-SQL preprint evaluating 500 queries with 73.4% result equivalence and 100% SQL execution success, alongside explainable anomaly-detection and high-throughput ML systems. Focused on LLM evaluation, system verification, and production-oriented ML engineering.

Skills

Languages: Python, SQL, JavaScript, TypeScript, Java

ML & AI: PyTorch, Scikit-learn, LightGBM, OpenCV, SHAP, NLP

LLM & Evaluation: RAG, FAISS, Sentence Transformers, LangChain, LangGraph, Text-to-SQL, SQLGlot, AST Verification, Benchmarking, Ablation Studies, RAGAS, PEFT (LoRA)

MLOps & Infrastructure: FastAPI, Docker, GitHub Actions, AWS (EC2, S3), Drift Detection, Model Monitoring, CI/CD

Databases & Tools: PostgreSQL, MongoDB, Redis, SQLite, Qdrant, Pinecone, Pandas, NumPy, SciPy

Work Experience

Wadhvani Foundation

Bangalore, India

AI Intern

Jun 2026 – Aug 2026

- Participated in architecture reviews for an enterprise-grade, multi-tenant AI platform, studying RAG retrieval, vector search, embeddings, and query routing across document-ingestion and conversational AI services.
- Traced Python/FastAPI workflows spanning OCR, LLM extraction, multilingual processing, embeddings, and Qdrant-based semantic retrieval.
- Studied embedding-model migration and retrieval architecture, analyzing how vector indexing and source-grounded RAG fit into the platform lifecycle.

Superhero Learning

Bangalore, India

Web Developer Intern

Jun 2025 – Sep 2025

- Deployed **8+** MERN pages serving **100+** active users, optimizing MongoDB indexing, query plans, and aggregation workflows to reduce API latency by **25%**.
- Improved load speed by **10%** and iteration time by **25%** through performance observability and asset compression.

Projects

Engineering Reliable LLM-Based Data Analysis – Research Preprint ^{DOI}

Python, SQL, RAG, SQLGlot

- Engineered a reliability-oriented Text-to-SQL pipeline combining graph-guided schema RAG, structured DAG planning, deterministic plan validation, AST-based SQL verification, and closed-loop repair.
- Evaluated **500 queries** across **8 e-commerce domains**, achieving **73.4% result equivalence**, **31.0% exact match**, and **100% SQL execution success** with **64.04s mean latency**.
- Conducted a controlled **100-query ablation** showing AST verification improved result equivalence from **15.0% to 26.0%** ($p = 0.0192$, $OR=3.75$).
- Audited **101 repair events**, identifying **22 false-positive regressions** versus **4 genuine recoveries**, motivating conservative, uncertainty-aware SQL verification.

SentinelGraph – Explainable Anomaly Detection Engine

Python, Streamlit, NetworkX, PyTorch, Docker

- Architected a **7-detector** behavioral anomaly ensemble for **1,000+ entities**, suppressing false positive alert rates by **>90%** via max-dominant risk fusion.
- Evaluated GNN and Graph-RAG traversal pipelines with RAGAS, achieving **0.942 PR-AUC** and **sub-10ms** scoring latency over **10K benchmark events**.
- Integrated counterfactual explainability and SHAP feature breakdowns, increasing policy explainability and threat triage efficiency by **30%**.

UnderdogAI – High-Throughput Event-Driven Decision Platform

FastAPI, Kafka, gRPC, Redis, PostgreSQL, Kubernetes

Kubernetes

- Benchmarked the HTTP API at **50 concurrent clients** and **1,000 requests**, increasing throughput **57%** from 352 to 553 req/s while reducing P50/P95/P99 latency by **37%/26%/29%** with zero failures.
- Built fault-tolerant Kafka processing with idempotent job handling, bounded retries, exponential backoff, and asynchronous workers; reduced server CPU time **42%**, peak RSS **10%**, and recovered two injected failures in **3.0s**.

Education

PES University

Bangalore, India

B.Tech., Computer Science (AI & Machine Learning)

Aug 2023 – May 2027

CGPA: 7.85

Coursework: Explainable AI, LLMs & Applications, Deep Learning, Reinforcement Learning, Machine Learning, Algorithms, Python